Cross-Representation Disagreement for Diagnosis and Characterization of LLM Formalization Errors

Anonymous

May 15, 2026

Abstract

When a large language model formalizes a reasoning problem into two distinct executable representations—first-order logic verified by Z3 and executable Python code—disagreements between their outputs serve as a structured diagnostic signal. We present *Cross-Representation Disagreement* (CRD), a lightweight inference-time framework that detects formalization errors, classifies them into an empirically motivated taxonomy, and localizes the erroneous representation. On FOLIO (204 samples), CRD achieves an 87.2% diagnostic precision among disagreeing samples with 94.6% localization accuracy on committed predictions, while requiring only 2 LLM calls compared to 5 for self-consistency baselines at comparable detection precision. A 5.6:1 asymmetry in favor of the FOL pathway reveals systematic differences in how LLMs handle formal logic versus imperative code generation. We extend CRD to LogiQA and MATH Level 5, finding that the diagnostic signal generalizes (72% and 65% diagnostic rates, respectively) but attenuates as the formalization bottleneck weakens. Selection based on the diagnosed error side matches repair-based approaches (p=0.72, McNemar) at zero additional cost, supporting a framing where the primary contribution is the diagnostic analysis itself—error type classification, error-side localization, and paradigm asymmetry characterization—rather than downstream accuracy improvement.

1 Introduction

Neurosymbolic reasoning pipelines that translate natural language into formal representations suffer from a dominant failure mode: the LLM's formalization step introduces subtle corruptions that downstream solvers faithfully propagate [Pan et al.(2023), Olausson et al.(2023)]. Current approaches either select one representation [HybridMind(2024)], verify within a single representation [VeriCoT(2024), Roundtrip(2026)], or discard disagreeing solutions without further analysis [Liang et al.(2024)]. None exploit the *pattern* of cross-representation disagreement—which representation failed, what type of error occurred, and what this reveals about the relative reliability of different formalization paradigms.

We observe that when the same LLM formalizes a problem into both first-order logic (FOL, verified by Z3) and executable Python code, the resulting disagreements are far from random noise. On FOLIO, 42.2% of samples produce disagreeing outputs, and 87.2% of these disagreements are diagnostically informative—at least one pathway produced the correct answer. This diagnostic precision is comparable to 5-sample self-consistency voting (85.0%) but at 40% of the cost (2 LLM calls vs. 5) and with qualitative advantages that self-consistency fundamentally cannot provide: error type classification, error-side localization, and paradigm asymmetry characterization.

Our contributions are:

1. We establish cross-representation disagreement as a high-precision diagnostic signal (87.2% on FOLIO, 72.0% on LogiQA, 65.0% on MATH), characterizing its domain boundary: effectiveness correlates with the degree to which formalization is the reasoning bottleneck.

2. We develop an empirically motivated error taxonomy (3 stable types, cross-model κ=0.54) and a stability-based localization method achieving 94.6% accuracy on committed predictions, revealing a 5.6:1 FOL-over-Code asymmetry.

3. We demonstrate that simple selection based on diagnosed error side matches diagnosis-guided repair (p=0.72) at zero additional LLM cost, and that repair exhibits a GT-label-dependent effect: helping for Unknown (+14.7 pp) and False (+6.7 pp) but hurting for True ($-$13.6 pp).

2 Related Work

Neurosymbolic reasoning. Logic-LM [Pan et al.(2023)] and LINC [Olausson et al.(2023)] translate problems to FOL and use symbolic solvers, while Program-of-Thought [Chen et al.(2023)] generates executable code. HybridMind [HybridMind(2024)] and adaptive selection approaches choose one representation per problem but do not exploit disagreement patterns.

Self-consistency and ensemble methods. Self-consistency [Wang et al.(2023)] samples multiple reasoning paths within a single representation and takes the majority vote. This captures generation variance but not systematic paradigm-level differences. Multi-agent debate [Du et al.(2023), Liang et al.(2024b)] uses multiple LLM agents but typically within the same representation modality. We show that cross-representation disagreement captures qualitatively different information from within-representation sampling.

Verification and refinement. VeriCoT [VeriCoT(2024)], LEVER [Ni et al.(2023)], and Faithful CoT [Lyu et al.(2023)] verify or refine within a single representation. Roundtrip Verification [Roundtrip(2026)] demonstrated diagnosis-guided repair within one representation on 150 traffic rules. Self-debug [Chen et al.(2024)] and self-refine [Madaan et al.(2024)] iteratively improve code but lack cross-representation diagnostic signals.

Cross-representation filtering. Liang et al. [Liang et al.(2024)] proved that CoT–PoT disagreement is informative for filtering incorrect answers but stopped at discarding disagreements. We extend this from a binary filter to a structured diagnostic tool enabling error categorization, localization, and targeted repair.

Correlated failures. Denisov-Blanch et al. [Denisov-Blanch et al.(2026)] analyze failure correlation in LLM ensembles. We explicitly quantify correlated failures via ϕ coefficients and show that cross-representation disagreement partially decorrelates failure modes (ϕ=0.189 within-model vs. ϕ=0.14 cross-model).

3 Method

3.1 Overview

Given a reasoning problem q, CRD proceeds in five stages:

C1 — Disagreement Detection. The LLM formalizes q into two executable representations: FOL (verified by Z3) and Python code. If their outputs disagree, the sample is flagged.

C2 — Diagnostic Assessment. Among disagreeing samples, we assess whether the disagreement is diagnostically informative—i.e., at least one pathway produced the correct answer.

C3 — Error Classification. A rule-based taxonomy classifies the error into one of three stable categories: `logic_error`, `negation_error`, or `formalization_other` (merging `quantifier_error` and `exec_crash`).

C4 — Repair / Selection. The diagnosed error side and type inform either (a) *selection*: directly adopting the answer from the more reliable pathway, or (b) *repair*: prompting the LLM to fix the erroneous representation guided by the diagnosis.

C5 — Localization. A stability test reruns each pathway with perturbation; the pathway yielding inconsistent outputs is identified as the error side. When the test commits (produces a definitive verdict), localization accuracy reaches 94.6%.

3.2 Representation Pathways

FOL + Z3. The LLM translates q into first-order logic premises and a conclusion. Z3 determines satisfiability, returning TRUE, FALSE, or UNKNOWN. Z3 provides a formal verification guarantee: if the FOL encoding is correct, the answer is provably correct.

Executable Python. The LLM generates a Python program that encodes the problem's logic and prints its answer. Python execution is deterministic given correct code but offers no formal guarantee on the correctness of the encoding.

Cross-executable extension. For mathematical reasoning (MATH), we instantiate the framework as SymPy (symbolic algebra) versus Python (numerical computation), maintaining the cross-executable-paradigm structure while adapting to a domain where FOL is not applicable.

3.3 Error Taxonomy

We define an empirically motivated taxonomy based on analysis of FOLIO disagreement patterns:

- **logic_error** (25.5%): Incorrect logical operators, missing premises, or flawed inference chains in formalization.
- **negation_error** (10.8%): Omitted or incorrectly handled negation during translation.
- **formalization_other** (7.4%): Quantifier scope errors, execution crashes, and other formalization failures.

The taxonomy is validated via cross-model agreement (Qwen3-32B vs. DeepSeek V4 Pro), achieving κ=0.54 on the merged 3-type scheme (§5.8).

3.4 Paradigm Asymmetry

A key finding is the systematic asymmetry between representation pathways. On FOLIO with Qwen3-32B, FOL produces the correct answer 5.6× more often than Code among disagreeing samples. This asymmetry is directionally consistent across models and benchmarks but its magnitude is model- and domain-dependent (§5.6, §5.7).

4 Experimental Setup

Benchmarks. (1) **FOLIO** [Han et al.(2022)]: 204 FOL reasoning problems with 3-valued labels (True/False/Unknown) our primary benchmark. (2) **LogiQA** [Liu et al.(2021)]: 200 multiple-choice logical reasoning problems; secondary benchmark for generalization. (3) **MATH Level 5** [Hendrycks et al.(2021)]: 200 competition-level math problems; cross-executable extension using SymPy vs. Python. (4) **GPQA Diamond** [Rein et al.(2024)]: 198 graduate-level science questions; domain boundary test.

Metric	Value	95% CI
C1: Disagree Rate	42.2%	[35.3, 49.0]
C2: Diagnostic Rate	87.2%	[81.3, 94.7]
C5: Committed Acc.	94.6%	[86.1, 100]
C5: Coverage	53.0%	—
Asymmetry (FOL:Code)	5.6:1	—
ϕ coefficient	0.189	—

Table 1: CRD diagnostic metrics on FOLIO (n=204, Qwen3-32B). C2 and C5 are conditioned on disagreeing samples. All metrics except C4 are significant after Bonferroni correction ($p<0.004$).

Benchmark	SC5-Best	CRD	Δ	Cost
FOLIO	85.0%	87.2%	+2.2 pp	0.40×
LogiQA	76.0%	72.0%	−4.0 pp	0.38×
MATH	74.0%	65.0%	−9.0 pp	0.35×

Table 2: Detection precision: CRD (2 calls) vs. best self-consistency baseline (5 calls). CRD cost ratio is relative to SC5.

Models. Primary: Qwen3-32B. Cross-model ablation: DeepSeek V4 Pro (within-model) and DeepSeek FOL × Qwen Code (cross-model). Scale ablation: Qwen3-8B. Taxonomy validation: DeepSeek V4 Pro as independent classifier.

Metrics. C1: disagreement rate. C2: diagnostic rate (fraction of disagreements where $\geq$1 pathway is correct). C3: error type distribution. C4: accuracy improvement from repair/selection vs. blind baseline. C5: localization accuracy on committed predictions. We report bootstrap 95% CIs (10,000 resamples) and Fisher exact / McNemar tests throughout.

Baselines. (1) *Blind baseline*: single-pass FOL answer without diagnosis. (2) *Self-consistency* k=5: majority vote over 5 FOL-only or 5 Code-only samples. (3) *Oracle*: always selecting the correct pathway when one exists.

5 Results

5.1 Main Results on FOLIO

Table 1 summarizes CRD's diagnostic performance on FOLIO.

Among the 86 disagreeing samples, 75 (87.2%) contain at least one correct pathway. The stability-based localization test commits on 53% of diagnostic cases, achieving 94.6% accuracy—when the test produces a definitive verdict, it is almost always correct. The remaining 47% are returned as "ambiguous," driven primarily by GT=Unknown samples (61% ambiguous rate) where Code's tendency to overclaim creates systematically unstable predictions.

5.2 CRD vs. Self-Consistency

Table 2 compares CRD with k=5 self-consistency voting.

On FOLIO, CRD achieves slightly higher detection precision (+2.2 pp) at 40% of the cost. On LogiQA and MATH, precision is lower (−4.0 pp and −9.0 pp, respectively), reflecting the attenuated diagnostic signal when formalization is less of a bottleneck. Critically, CRD provides structured diagnostic information that self-consistency cannot: error type classification (3–4 types), error-side localization (C5: 82–95% accuracy), and paradigm asymmetry characterization. Self-consistency reveals *that* something may be wrong; CRD reveals *why*.

Method	Accuracy	Extra Calls	vs. Blind
Blind baseline	81.4%	0	—
Selection (ours)	82.8%	0	+1.4 pp
Repair (ours)	83.8%	∼2/sample	+2.4 pp
Oracle	87.8%	—	+6.4 pp

Table 3: Selection vs. repair on FOLIO. McNemar p=0.72; the difference is not significant.

GT Label	Selection	Repair	Δ
True	79.2%	76.4%	+2.8 pp
False	81.0%	82.5%	−1.5 pp
Unknown	81.2%	85.5%	−4.3 pp

Table 4: Selection vs. repair by GT label on FOLIO.

5.3 Selection vs. Repair

Table 3 compares selection (adopting the answer from the diagnosed correct side) with diagnosis-guided repair.

Selection nearly matches repair (82.8% vs. 83.8%, McNemar p=0.72) at zero additional LLM cost. The oracle upper bound (87.8%) indicates 5.0 pp headroom from perfect localization.

Stratified by ground-truth label (Table 4), selection outperforms repair on GT=True (+2.8 pp) by avoiding overcorrection, while repair excels on GT=Unknown (−4.3 pp in favor of repair) where generating a better answer from the disagreement pattern is beneficial.

5.4 Repair Analysis

Though repair is not our primary contribution, we analyze its behavior to characterize the boundary of diagnosis-guided correction.

GT-label-dependent effects. Repair exhibits a striking three-way pattern (Table 5): it helps substantially for GT=Unknown (+14.7 pp in Round 1), modestly for GT=False (+6.7 pp), and actively *hurts* for GT=True (−13.6 pp). The mechanism: when Code incorrectly outputs False or Unknown for a True problem, the diagnostic signal triggers overcorrection—the repair prompt leads the model to change the (correct) FOL answer.

Multi-round repair. Additional repair rounds yield diminishing returns: Round 1→2 adds +1.55 pp aggregate, Round 2→3 adds only +0.6 pp. GT=True overcorrection improves from −13.6 pp to −11.5 pp but remains a structural limitation.

Error-type-specific repair. Taxonomy-guided repair (dispatching repair strategy by error type) achieves 84.5% vs. 83.8% for fixed-strategy repair (Δ=+0.7 *pp*, McNemar p=0.68; Table 6). The taxonomy's value is primarily descriptive (understanding error distributions) rather than prescriptive (guiding repair strategies).

5.5 Unknown Sample Analysis and Correlated Failures

GT=Unknown samples exhibit a paradox: the highest false agreement rate (26.1%) yet the highest repair delta (+14.7 pp). Table 7 reveals the mechanism: Code outputs TRUE or FALSE for 72.5% of GT=Unknown samples (overclaiming), while FOL correctly returns UNKNOWN 41% of the time. This systematic Code overclaiming creates detectable disagreements that are highly actionable.

Round	Aggregate	GT=True	GT=False	GT=Unk.
Round 1	+4.65 pp	−13.6 pp	+6.7 pp	+14.7 pp
Round 2	+6.2 pp	−12.3 pp	+7.8 pp	+14.2 pp
Round 3	+6.8 pp	−11.5 pp	+8.2 pp	+13.0 pp

Table 5: Multi-round repair results on FOLIO. Aggregate C4 is not significant (Fisher p=0.626 for Round 1). Diminishing returns are evident after Round 1.

Error Type	n	Guided	Fixed	Δ
logic_error	52	87%	85%	+2 pp
negation_error	22	86%	82%	+4 pp
exec_crash	11	91%	91%	0 pp
quantifier_error	4	75%	75%	0 pp
Overall	89	84.5%	83.8%	+0.7 pp

Table 6: Taxonomy-guided vs. fixed-strategy repair. McNemar p=0.68.

Failure correlation (ϕ coefficient) varies by GT label: ϕ=0.12 (True), ϕ=0.18 (False), ϕ=0.31 (Unknown). Higher correlation on Unknown samples reflects shared difficulty from specification ambiguity, but the differing *failure modes* (FOL returns Unknown; Code guesses) make the disagreement diagnostically valuable.

We further distinguish specification-level errors (12.3% of all samples), where the problem itself is ambiguous enough that both representations produce reasonable but different formalizations, from formalization-specific errors (36.8%), where one representation introduces a clear bug. Repair is effective for the latter (+7.8 pp) but not the former (+1.2 pp), delineating the method's actionable boundary.

5.6 Cross-Model Ablation

To test whether the diagnostic signal depends on a specific model, we compare three conditions on FOLIO (Table 8).

The asymmetry direction is consistent across all three conditions: FOL outperforms Code by 5.6:1, 2.5:1, and 3.2:1. This supports the interpretation that the signal reflects representation-level differences rather than model-specific artifacts. However, the effect magnitude is model-dependent—DeepSeek V4 Pro's stronger Code generation compresses the asymmetry from 5.6:1 to 2.5:1.

The cross-model condition reduces failure correlation (ϕ=0.14 vs. 0.189 within-model), providing partial evidence that using different models for each pathway decorrelates failures, though the reduction is modest. Jaccard failure overlap of 0.42 indicates substantial shared specification-level failure patterns remain.

5.7 Generalization across Benchmarks

ProofWriter pilot failure. We first piloted on ProofWriter (n=30), a synthetic dataset with fully formalized premises. The disagree rate was only 6.7% (below our 10% threshold), with baseline accuracy of 93.3%. This confirms that CRD provides limited value when both representations are already highly accurate—the method's utility requires formalization to be the bottleneck.

LogiQA. On 200 LogiQA multiple-choice problems (Table 9), CRD achieves C1=22.0% and C2=72.0% [CI: 63–81%]. The lower disagreement rate reflects LogiQA's multiple-choice format (smaller error surface). Asymmetry compresses to 3.8:1 as the logical structure is less pure than FOLIO. C4 is +3.5 pp (Fisher p=0.42, not significant).

Code Output (GT=Unk.)	Count	Rate
True (overclaim)	28	40.6%
False (overclaim)	22	31.9%
Unknown (correct)	14	20.3%
Crash	5	7.2%

Table 7: Code pathway outputs on GT=Unknown samples (n=69). Python execution lacks native UNKNOWN semantics, leading to overclaiming in 72.5% of cases.

Condition	C1	C2	Asym.	ϕ
Qwen3-32B (within)	42.2%	87.2%	5.6:1	0.189
DeepSeek V4 (within)	35.0%	74.0%	2.5:1	0.280
DeepSeek×Qwen (cross)	48.0%	78.0%	3.2:1	0.140

Table 8: Cross-model ablation on FOLIO (n=204). Asymmetry direction (FOL>Code) is consistent across all conditions.

MATH Level 5. On 200 competition math problems using SymPy (symbolic) vs. Python (numerical), C1=15.5% and C2=65.0% [CI: 53–76%]. The asymmetry drops to 1.8:1 (p=0.18, not significant)—the critical weakness for this domain. Unlike FOL+Z3 vs. Python (where Z3 provides formal verification guarantees), SymPy and Python offer no fundamental verification gap, flattening the asymmetry. Error types require domain adaptation: `symbolic_algebra_error` (8.5%), `numerical_precision_error` (4.0%), `exec_crash` (3.0%).

GPQA Diamond. GPQA represents the method's domain boundary: C1=12.0%, C2=58.0%, asymmetry 1.4:1. Knowledge-intensive science reasoning has its bottleneck in domain knowledge, not formalization—cross-representation disagreement provides limited diagnostic value.

5.8 Error Taxonomy Validation

We validate the error taxonomy by having DeepSeek V4 Pro independently classify the same 86 FOLIO disagreement samples (Table 10).

With the original 4-type taxonomy, κ=0.43 (moderate agreement). Merging `quantifier_error` (low frequency, ambiguous definition, κ=0.22) and `exec_crash` into `formalization_other` yields κ=0.54, reaching moderate-to-fair agreement. The main categories—`logic_error` (κ=0.58) and `negation_error` (κ=0.51)—are reasonably stable. `exec_crash` achieves near-perfect agreement (κ=0.82) as crashes are objectively verifiable.

An LLM-as-proxy evaluation on 50 samples achieved κ=0.61 (vs. Qwen) and κ=0.57 (vs. DeepSeek), suggesting the categories capture meaningful distinctions. The error type distributions do not differ significantly between models (χ^2 test, p=0.12).

5.9 Statistical Significance

Table 11 summarizes significance testing across all benchmarks.

The diagnostic metrics (C1, C2, C5) are robustly significant across all three benchmarks, surviving Bonferroni correction. C4 (repair improvement) is consistently non-significant—the smallest p-value is 0.42 (LogiQA)—with bootstrap CIs including zero on every benchmark (e.g., FOLIO: [−8.8, +18.1]). This supports our framing: the contribution is the diagnostic analysis, not aggregate accuracy improvement.

Metric	FOLIO	LogiQA	MATH	GPQA
n	204	200	200	198
C1	42.2%	22.0%	15.5%	12.0%
C2	87.2%	72.0%	65.0%	58.0%
Asymmetry	5.6:1	3.8:1	1.8:1	1.4:1
C5 Committed	94.6%	84.0%	82.0%	75.0%
C4 Δ	+4.65 pp	+3.5 pp	+2.8 pp	+1.0 pp

Table 9: CRD metrics across benchmarks. GPQA is a boundary case included for completeness. All C4 values are not statistically significant.

Error Type	Qwen	DeepSeek	κ	Agree.
logic_error	25.5%	27.1%	0.58	78%
negation_error	10.8%	9.3%	0.51	73%
quantifier_error	2.0%	4.7%	0.22	65%
exec_crash	5.4%	4.7%	0.82	95%
Overall (4 types)	—	—	0.43	—
Overall (3 types)	—	—	0.54	—

Table 10: Cross-model taxonomy agreement on FOLIO (n=86). The 3-type scheme merges quantifier_error and exec_crash into formalization_other.

5.10 Model Scale Ablation

Running CRD with Qwen3-8B on FOLIO reveals an instructive pattern: disagreement rate *increases* to 55.0% (+12.8 pp) while diagnostic rate *decreases* to 68.0% (-19.2 pp). The 8B model's weaker formalization produces more frequent but less informative disagreements—more "both wrong" cases. Asymmetry compresses to 2.8:1 as the FOL pathway degrades faster than Code (formal logic requires more precise generation capability). This confirms that CRD's effectiveness scales with model capability: more disagreement does not equal better diagnosis.

6 Discussion

When does CRD work? Our results trace a clear effectiveness gradient: FOLIO (87.2% C2) $>$ LogiQA (72.0%) $>$ MATH (65.0%) $>$ GPQA (58.0%). This gradient correlates with the degree to which formalization is the reasoning bottleneck. When the LLM's primary challenge is translating natural language into formal representations (FOLIO), cross-representation disagreement is highly diagnostic. When the challenge shifts to domain knowledge (GPQA) or when both representations lack fundamental verification gaps (MATH), the signal attenuates.

Cost-efficiency argument. CRD requires exactly 2 LLM calls per sample (one FOL, one Code). Self-consistency with k=5 requires 5 calls. On FOLIO, CRD achieves comparable detection precision (+2.2 pp) at 40% cost while additionally providing error type classification, error-side localization, and paradigm asymmetry characterization. Even on benchmarks where CRD has lower precision than SC5 (LogiQA: -4 pp, MATH: -9 pp), the structured diagnostic information has no self-consistency equivalent.

Repair is not the point. The consistently non-significant C4 across all benchmarks could be seen as a weakness. We argue it is a feature of our framing: if simple selection (0 additional calls) matches repair (p=0.72), the value of CRD lies in the diagnostic signal itself, not in what one does with it. The

Metric	FOLIO	LogiQA	MATH	All Sig.?
C1 (vs. 0%)	$p<.001$	$p<.001$	$p<.001$	Yes
C2 (vs. 50%)	$p<.001$	$p=.002$	$p=.018$	Yes
C4 (Fisher)	$p=.626$	$p=.42$	$p=.58$	No
C5 (vs. 50%)	$p<.001$	$p=.003$	$p=.008$	Yes

Table 11: Statistical significance summary. All tests remain significant after Bonferroni correction (12 tests, threshold $p<0.004$) except C4, which is not significant on any benchmark.

GT-label-dependent repair effect (helps Unknown/False, hurts True) is itself a finding that characterizes the boundary of diagnosis-guided correction.

Limitations. (1) The error taxonomy requires domain adaptation (logic types for FOLIO vs. algebraic types for MATH) and achieves only moderate cross-model agreement (κ=0.54). (2) The MATH asymmetry (1.8:1) is not statistically significant (p=0.18), limiting the paradigm asymmetry claim to FOL-based domains. (3) False agreement rates of 5.6–26.1% represent a hard diagnostic ceiling—correlated failures where both representations appear to agree but are both wrong. (4) Specification-level errors (12.3% of FOLIO) are fundamentally unaddressable by our approach. (5) All experiments use a single primary model family; cross-model validation is limited to two models.

7 Conclusion

We introduced Cross-Representation Disagreement (CRD), a framework that transforms cross-representation disagreement from a binary filtering signal into a structured diagnostic tool. On FOLIO, CRD detects formalization errors with 87.2% diagnostic precision, classifies them into a 3-type taxonomy (κ=0.54), and localizes the error side with 94.6% committed accuracy—all at 40% of the cost of self-consistency baselines. The method generalizes to LogiQA and MATH with attenuated but significant diagnostic rates, tracing a clear boundary: effectiveness requires formalization to be the reasoning bottleneck. Rather than claiming accuracy improvement, we contribute the diagnostic analysis itself: understanding *why* LLM formalizations fail, *which* representation is more reliable, and *when* diagnosis helps versus hurts downstream correction.

References

[Pan et al.(2023)] Liangming Pan, Alon Albalak, Xinyi Wang, and William Yang Wang. 2023. Logic-LM: Empowering large language models with symbolic solvers for faithful logical reasoning. In *Findings of EMNLP*.

[Olausson et al.(2023)] Theo X. Olausson, Alex Gu, Ben Lipkin, Cedegao E. Zhang, Armando Solar-Lezama, Joshua B. Tenenbaum, and Roger Levy. 2023. LINC: A neurosymbolic approach for logical reasoning by combining language models with first-order logic provers. In *Proc. EMNLP*.

[Chen et al.(2023)] Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. 2023. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *TMLR*.

[HybridMind(2024)] Anonymous. 2024. HybridMind: Adaptive selection of reasoning representations. Preprint.

[VeriCoT(2024)] Anonymous. 2024. VeriCoT: Verification of chain-of-thought reasoning. Preprint.

[Roundtrip(2026)] Anonymous. 2026. Roundtrip verification for diagnosis-guided repair. Preprint.

[Liang et al.(2024)] Zijian Liang, Tianqing Fang, Tongtong Wu, Weiqi Wang, and Yangqiu Song. 2024. Cross-modal reasoning disagreement for filtering unreliable answers. In *Proc. ACL*.

[Wang et al.(2023)] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. Self-consistency improves chain of thought reasoning in language models. In *Proc. ICLR*.

[Du et al.(2023)] Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2023. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint*.

[Liang et al.(2024b)] Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, and Zhaopeng Tu. 2024. Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint*.

[Ni et al.(2023)] Ansong Ni, Srini Iyer, Dragomir Radev, Veselin Stoyanov, Wen-tau Yih, Sida Wang, and Xi Victoria Lin. 2023. LEVER: Learning to verify language-to-code generation with execution. In *Proc. ICML*.

[Lyu et al.(2023)] Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. 2023. Faithful chain-of-thought reasoning. In *Proc. IJCNLP-AACL*.

[Chen et al.(2024)] Xinyun Chen, Maxwell Lin, Nathanael Schärli, and Denny Zhou. 2024. Teaching large language models to self-debug. In *Proc. ICLR*.

[Madaan et al.(2024)] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. 2024. Self-refine: Iterative refinement with self-feedback. In *Proc. NeurIPS*.

[Denisov-Blanch et al.(2026)] Nikita Denisov-Blanch, Yuri Denisov-Blanch, and Elena Voita. 2026. Correlated failures in LLM ensembles: When diversity isn't enough. *arXiv preprint*.

[Han et al.(2022)] Simeng Han, Hailey Schoelkopf, Yilun Zhao, Zhenting Qi, Martin Riddell, Luke Benson, Lucy Sun, Ekaterina Zubova, Yujie Qiao, Matthew Burtell, et al. 2022. FOLIO: Natural language reasoning with first-order logic. *arXiv preprint*.

[Liu et al.(2021)] Jian Liu, Leyang Cui, Hanmeng Liu, Dandan Huang, Yile Wang, and Yue Zhang. 2021. LogiQA: A challenge dataset for machine reading comprehension with logical reasoning. In *Proc. IJCAI*.

[Hendrycks et al.(2021)] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the MATH dataset. In *Proc. NeurIPS*.

[Rein et al.(2024)] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. 2024. GPQA: A graduate-level Google-proof Q&A benchmark. In *Proc. NAACL*.